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Option Theory Project

Mathematical Modelling of Stochastic Systems

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1 Introduction to options

In this study, we seek to model financial systems and tools mathematically. To this end, we have selected option contracts as the financial tool of examination.

Option contracts are legally binding agreements between the writer and buyer. The buyer obtains the right, but not the obligation, to purchase or sell the underlying asset at a pre-determined price, known as the strike price, within a designated period of time referred to as the term. There are two types of option contracts: call options, which grant the buyer the right to buy shares, and put options, which give the buyer the right to sell shares.

The trading of options occurs on an options exchange, where the contracts are standardized in terms of size, term, and expiration date. Typically, a stock option contract constitutes 100 shares, and the term is commonly 3, 6, 9, or 12 months. The expiration date is always the Saturday after the third Friday of the expiry month, unless the exchange is closed on said Friday.

A differentiation is made between European and American options. European options may only be exercised on the specified expiration date, while American options may be exercised at any point prior to the expiration date. In this paper we focus on the former.

When the buyer acquires an option, they possess the right to purchase or sell shares at the strike price but are not obliged to do so. The buyer's risk is limited to the premium paid, along with any commission fees imposed by the bank or broker that facilitated the transaction. If the price of the underlying asset fails to meet the buyer's expectations they may elect not to exercise their right, incurring a loss equivalent to the premium paid and commission fees.

The writer of the option is the counter party of the option buyer and is obliged to deliver or purchase the specified number of shares at the strike price if the option buyer exercises their right. In return, the writer receives a premium. On the options exchange, individuals have the capacity to not only buy and sell options, but also write options. The exchange establishes numerous strike prices for each underlying asset based, of course, on the current share price and often closely resembling it. For instance, if the price of share ABC is 80, options with strike prices of 75, 80, and 85 are commonly created. In the event of significant fluctuations in the price of the underlying asset, new options series with adjusted strike prices will be introduced, while the original series will remain listed until their expiration date, enabling continued trading opportunities for investors.

2 The price of an option

In this section we lay the groundwork for what the price of an option really means, namely the premium that a buyer of an option pays to the writer of an option as compensation for the latter's risk associated with the promise to deliver or purchase a predetermined number of shares. This premium, which is the option price, depends on various variables, including the current share price (S), the strike price (K), the interest rate (r), the remaining time until expiration (t), the volatility of the share (σ), and the height of the dividend (D).

We will examine the impact of these variables in turn. Firstly, we consider the variables S and K. For a call option, the price of the option (C) is a function of the current share price (S) and the strike price (K). Since the share price fluctuates the price of a certain option is hard to determine, as the value is tied to share price at the expiration date, which can of course be anything. This paper illustrates the derivation of the first model made to estimate the price (i.e. the value) of options.

At expiration, we denote the share price by S^* and the call price by C^*. This latter term can easily be determined to be

$$C^* = \max\{0, S^* - K\}.$$

This is because the call option provides the right to purchase shares at the strike price of K at the expiration date. If $S^* > K$, it is advantageous to exercise this right, as the option owner would purchase the share at a price lower than its actual value and then sell the shares at the current rate of S^* for a profit. If $S^* \leq K$ the call option is worth nothing as it gives the right to buy shares for more than they're worth. This makes it worthless, and it is then of course better to let it expire.

When considering the price of a call option at a time prior to the expiration date, we can only obtain a lower bound for the option price. We must also differentiate between American and European options. For an American call option, we have

$$C \geq \max\{0, S - K\}.$$

This is because the option provides the right to purchase shares at the strike price at any time prior to expiration. The value $\max\{0, S - K\}$ is referred to as the intrinsic value of the call option. The difference between C and the intrinsic value is referred to as the time value of the option. Note that, like C and S, the intrinsic value and time

value also depend on time (as, for example, the time value decreases as the expiration date approaches).

Similarly, the intrinsic value of an American put option is the difference between the strike price and the actual price of the underlying asset, $K - S$. An American put option is a financial tool which is not obligatory to exercise. Because of this, if $K - S$ is negative the owner of said option is better off not exercising the option meaning it gives no value and is worth 0. In mathematical terms, $P \geq \max\{0, K - S\}$, where the inequality arises from the fact that $K - S$ might be bigger in the future[1].

For the price of a put option P^* at expiration the reasoning is analogous. At expiration, $P^* = \max\{0, K - S^*\}$ and can again not be negative for the same reasons as the call option. Note that the K and S^* are flipped when compared to C^*, as now the strike price will be higher than the share price (if not the price is 0).

We can combine the formulas for call and put options at expiration to observe that $P^* + C^* = |S^* - K|$.[2] This is because if $K < S^*$, $C^* = \max\{0, S^* - K\} = S^* - K$ and $P^* = \max\{0, K - S^*\} = 0$. On the other hand, if $K > S^*$, $P^* = \max\{0, K - S^*\} = K - S^*$ and $C^* = \max\{0, S^* - K\} = 0$. This means that regardless of whether $K > S^*$ or $S^* > K$, we have $P^* + C^* = |S^* - K|$.

The relationship between the value of an option and the prevailing interest rate r has not yet been considered. To simplify the analysis, we will assume that options are only exercised at the expiration date, and therefore limit our discussion to European options. To clarify the concept of interest rate, it is important to note that the interest earned on a bank account is dependent on the account's terms and conditions. For example, a time deposit with a longer term will typically yield a higher interest rate than an account that allows for immediate withdrawal.

It is important to distinguish between annual interest rates and continuous interest rates. As an example, consider an initial deposit of 100 in a savings account with a yearly interest rate of 6%, meaning that after a year the credit would be $100 \cdot 1.06 = 106$. If another bank gives an interest rate of 3% every 6 months, the credit after a year would be $100 \cdot 1.03 \cdot 1.03 = 106.09$, slightly more than the first bank. This method of calculating interest is called compound interest.

To better visualize and understand compound interest we build on the illustration above. Consider the situation where bank n gives $\frac{6}{n}\%$ interest n times a year over the

[1] Q2.1a)
[2] Q2.1b)

amount in the account at that time. The resulting amount c_n after one year of an investment of 100 would be

$$c_n = 100\left(1 + \frac{6}{100n}\right)^n,$$

where the first factor of 100 is the initial investment, and the second a consequence of calculating with percentages. In the compound interest rate, $\frac{6}{n}$ is divided by 100 and summed to 1 to result in the interest rate factor for the first interval of compounded interest. Thus the factor of $(1 + \frac{6}{100 \cdot n})$ is raised to the power of n to finally result in the full amount received after compounding the interest on the initial amount n times, thus the resulting formula for the return of an investment of 100 to bank n. [3]

To continue with the discussion, we first claim the following result

Theorem 2.1. *The sequence* $\left(1 + \frac{r}{n}\right)^n$ *is increasing with respect to* n*, where* r *is a fixed real number.*

See Appendix A.1 for the proof of the theorem. In the theorem, the variable r stands for the interest rate, which in our case is 3/50. We can apply this theorem to our formula then. The conceptual meaning of the theorem is that if bank n gives $(6/n)\%$ interest n times a year over the amount in the account at that time, then the bigger the n, the bigger the return on investment for the liquidity provider. The next step is therefore clear. If the sequence is increasing, to better understand the behaviour of such interest, we want to find the limit as n goes to infinity of the compound interest formula, for which we use the following result

Theorem 2.2. *The limit of the function* $(1 + \frac{r}{n})^n$ *as* $n \to \infty$ *is equal to* e^r.

See Appendix A.2 for the proof of the theorem. Using this result, we can see that $100 \cdot \lim_{n\to\infty}(1 + \frac{6}{100 \cdot n})^n = 100 \cdot e^{\frac{6}{100}} \approx 106.1836$. So 106.1836 is approximately the maximum amount one could earn after a year by investing 100 in a bank that gives interest every infinitesimal fraction of a second.

We then adopt the convention that if the interest rate is equal to r, then for a deposit of c_0 at time $t = 0$, an amount of $c_0 e^{rt}$ can be claimed after a time t (measured in years). This comes from the fact that the generalized version of the compound interest formula is given by

$$c_n = c_0\left(1 + \frac{r}{n}\right)^{nt} \to c_0 e^{rt} \text{ as } n \to \infty,$$

with c_0 the initial amount invested. The interest rate r is known as the risk-free interest rate, the interest rate that theoretically belongs to an investment with no risk (we will

[3] Q2.2

say more about the real-world situation later). Roughly, we consider it to be "risk-free" since the money is not invested into any asset on the market whose price can fluctuate and therefore depreciate (here comes the risk), but it is money lent as credit and therefore "guaranteed" to be returned at the agreed interest rate to the borrower.

We are considering the call option price C (or put option price P) at a specific time, denoted by the variable t, which represents the remaining term until expiration. The economic principle of arbitrage states that

> It should not be possible to achieve a guaranteed profit without risk or prior investment.

Because of this principle we can prove relations between our variables C and P through contradiction. We assume the contrary is true and try to find a strategy to get a guaranteed profit (i.e. no risk).

Let us apply this principle to show that for an European call option, we have

$$C \geq S - Ke^{-rt}.$$

Suppose $C < S - Ke^{-rt}$, then an individual can buy the call option at the price C, sell the share at the price S and deposit the amount of Ke^{-rt} in an account with interest rate r, yielding a net amount of $S - Ke^{-rt} - C > 0$. At expiration, if $S^* \geq K$, the call can be exercised buying back the share at the price of K euro. This can be financed using the amount in the savings account which by now increased to exactly K euro. If $S^* < K$, let the call expire and buy the share on the exchange for S euro. This can again be financed using the amount of K euro in the savings account; the additional profit is $K - S$ euro.

The arbitrage principle also applies to European put options, deducing the lower bound

$$P \geq Ke^{rt} - S.$$

We again assume the contrary, that is, $P < Ke^{rt} - S$. One can always achieve a profit with the following strategy. Buy the put option at the price P, buy the share at the price S, and finance all this with the money obtained by borrowing the amount of Ke^{rt} euro from an account with interest rate r. This yields a net amount of $Ke^{rt} - S - P > 0$ euro. At the expiration date, there are two possibilities: $S^* < K$ or $S^* \geq K$. In the first case, the put can be exercised selling back the share at the price of K euro, and using this money to give the borrowed money back. The final net balance is thus $Ke^{rt} - S - P > 0$

euro. If $S^* \geq K$, let the put expire and sell the share on the exchange for S^* euro. You have the same balance plus an additional profit of $S^* - K$ euro[4].

Following the results, the conclusion that C is an increasing function and P is a decreasing function of r is reasonable. To examine the impact of volatility on the call option, we consider a scenario where the share price is below the strike price at a specific time before the expiration date, resulting in a zero intrinsic value for the call. However, this does not mean that $C = 0$, only that the value of the call is solely based on its time value. A higher volatility results in more opportunities for the share price to rise above K, increasing the value of the call.

The reader versed in economic theory will point out our disregard for dividends, which we purposefully ignore for sake of clarity of the argument. A dividend is the distribution of a portion of a company's earnings to its shareholders in the form of stock or cash payments. Dividends can have an impact on the value of options contracts, as they can affect the underlying stock price. If we were to consider them, a higher dividend would result in a decreased value of call options and increased value of put options. This is because the share goes "ex-dividend" on the day of dividend payment, decreasing the share price by the amount of the dividend.

In the case of a call option on a non-dividend paying share, it is never advantageous for the investor to exercise the right to buy before expiration. Instead, it is better for the investor to hold the option instead of the underlying share or sell the option instead of the underlying share, as the option's price will always be higher than its intrinsic value. Thus, for non-dividend paying shares, the prices of American and European call options are the same. On the other hand, exercising the right to sell before expiration may be profitable in the case of a put option, leading to a slightly higher price for an American put option compared to a European put option with the same expiration date and strike price. In the next section, we will analyze European options. We close this section off with the following table indicating the influence of all of our variables.

[4] Q2.3

Variable	Factor	P	C
S	current share price	$\downarrow$	$\uparrow$
K	strike price	$\uparrow$	$\downarrow$
r	risk-free interest rate	$\downarrow$	$\uparrow$
t	remaining term	$\uparrow$	$\uparrow$
σ	volatility	$\uparrow$	$\uparrow$
D	dividend	$\uparrow$	$\downarrow$

Table 1: Effect of determining factors on the option price

3 A binomial model for option prices

In this section we delve into the relationship between the price of an option and the price of the underlying share. Our analysis will be conducted in two stages. First, we consider a single trading period assuming that it is the final trading period before the expiration of the option. Second, we expand our analysis to include multiple trading periods. For clarity, we initially only consider call options and make several assumptions: a constant interest rate, no dividends being paid during the option's term, and we discard taxes and commissions.

To illustrate the relationship between the price of the option and the price of the underlying share, we introduce the concept of a portfolio and denote its value by T. A portfolio consists of a quantity Δ of shares (not necessarily full shares) and a sum of money B in an interest-bearing account. Here, B can also be negative, namely if the portfolio holder has a debt because of an outstanding loan. The value of the portfolio is dynamic and can change over time due to fluctuations in the share price and the interest earned on the money in the account.

We first analyze a simple scenario in which trading takes place at specific times and there is only one more trading period before the option expires. Let the option price at this moment be denoted by C, and the share price by S. The interest rate over one trading period is assumed to be $\hat{r}$, and we define ρ as $1 + \hat{r}$. During the trading period, we assume that the share price can either increase to $u \cdot S$ or decrease to $d \cdot S$, with the probability of the share price rising denoted by q, and the share price dropping with probability $1 - q$.

Next, we examine the price development of the call option C and portfolio T over the single trading period. At the end of the trading period there are two possibilities for the

portfolio

$$\begin{cases} T_u = uS\Delta + \rho B, & \text{with probability } q \\ T_d = dS\Delta + \rho B, & \text{with probability } 1-q \end{cases}$$

As for the call option, there are also two possibilities

$$\begin{cases} C_u = \max\{0, uS - K\}, & \text{with probability } q \\ C_d = \max\{0, dS - K\}, & \text{with probability } 1-q \end{cases}$$

As an observation, note that by selecting the values of Δ and B appropriately, we can guarantee, even if we do not know the value of the probability q, that the value of the portfolio after the one trading period is equal to the price of the call option. This unfortunately is only possible if we make the simplifying assumption of the posibility of buying fractional shares.

Since we want to get the values of Δ and B that guarantee that the value of the portfolio is equal to the price of the call option after the one trading period, we combine the outcomes with the same probability to get a system of two equations. The system looks like and is solved as follows [5]

$$\begin{cases} C_u = uS\Delta + \rho B \\ C_d = dS\Delta + \rho B \end{cases} \implies \begin{cases} \Delta = \frac{C_u - C_d}{S(u-d)} \\ B = \rho^{-1}\left(C_u - \frac{u(Cu - C_d)}{u-d}\right) \end{cases}. \tag{1}$$

With all the ground prepared, we state the following claim we wanted to get to.

<u>Claim:</u> If we choose Δ and B to be as as solved in Equation 1, then one period before expiration the value of our portfolio will also be equal to the price of the call option, that is, $C = S\Delta + B$. [6]

See proof of the claim in Appendix A.2. Now that we got a first pricing expression for the the call option, we want to remodel it to make it more understandable. Notice that, using the values we obtained for Δ and B, we have

$$C = S\Delta + B = S\frac{C_u - C_d}{S(u-d)} + \frac{1}{\rho}(C_u - uS\Delta) = \frac{C_u - C_d}{u-d} + \frac{1}{\rho}\left(C_u - u\frac{C_u - C_d}{u-d}\right) \tag{2}$$

$$= \frac{1}{\rho}\left(\frac{\rho C_u - \rho C_d - dC_u - uC_d}{u-d}\right) = \rho^{-1}\left(\frac{\rho - d}{u-d}C_u + \frac{u-\rho}{u-d}C_d\right) = \rho^{-1}[pC_u + (1-p)C_d].$$

Note in the last step we introduced the variable $p = \frac{\rho - d}{u-d}$. Thus we are able to define an equation for the price of a call option after one trading period, which only depends on

[5] Q3.1
[6] Q3.2

u, d, and C. The value of $p = \frac{\rho - d}{u - d}$ can be interpreted as the value that q would take if the expected value of the share after the trading period was equal to what could be achieved from an investment of S in a risk-free account. In other words, p is the value that solves for x in the equation $xuS + (1 - x)dS = \rho S$. On the RHS, ρ represents the amount that would be achieved from an investment of S in a risk-free account with interest rate $\hat{r}$, where $\rho = 1 + \hat{r}$. We can see that the LHS is the expected value of the share after the trading period if x were the probability of the share rising by a factor of u, and thus $1 - x$ being the probability of S decreasing by a factor d. We can see that plugging in $p = \frac{\rho - d}{u - d}$ for x, that p is the solution of the equation, so this value may indeed be interpreted as the value that q would take if the expected value of the share after the trading period was equal to what could be achieved from an investment of S in a risk-free account.

Interestingly, the actual probability of the share price rising or falling has no effect on the option price. The arbitrage principle has a greater influence and dictates the option price. If this principle is violated, one can potentially earn unlimited money without any risk, which is not possible in a rational market. To expand our analysis and better illustrate and understand the possible changes in the price of a call option and its replicating portfolio, we examine the call option with two remaining trading periods before expiration. We've illustrated all possible fluctuations for a specific set of variables in Figure 1.

The share price S can either increase by a factor of u or decrease by a factor of d both times. In the development of this diagram, for the sake of simplicity, we assume no interest rate, that is, $\rho = 1$. Also, suppose $S = 36$, $K = 30$, $u = 3/2$ and $d = 2/3$. With all these assumptions, and knowing only the current share price S, we can already price the current call option, using the binomial model and using $C = \rho^{-1}[pC_u + (1 - p)C_d]$ by working from right to left. Results are presented in Figure 1.

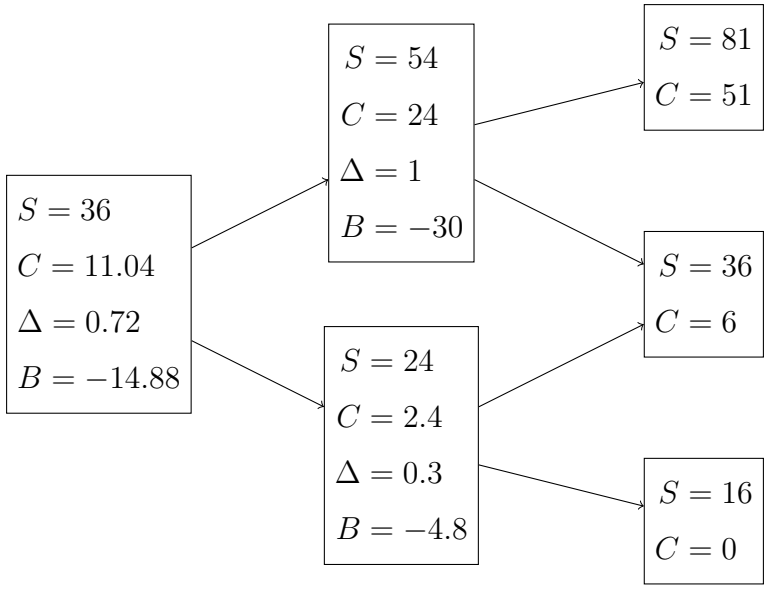

Figure 1: Binary tree of call option possibilities [7]

Note that since the rightmost column is at expiration, we can get the call option price by its definition at expiration, namely $C^* = \max\{0, S^* - K\}$. Therefore, we start at the top entry in the middle column and apply the formula $C = \rho^{-1}[pC_u + (1-p)C_d]$ to get the call option price. We do this for the one below as well and then finally on the leftmost, using the just obtained data for the option prices in the middle column and newly the aforementioned formula. Note that we also calculated Δ and B using Equation 1 for every possible case. It is left to the reader to check that $C = S\Delta + B$ at every given possible case, as we previously explained it must be the case.

Let us now write C_{uu} for the value of the call option two periods from now if the price of the underlying share has gone up twice; and let us define C_{ud}, C_{du}, and C_{dd} analogously. Observe that $C_{ud} = C_{du}$. Thus we can use the previously deduced equation $C = \rho^{-1}[pC_u + (1-p)C_d]$ to deduce the new version of the same equation now considering one extra time frame pre-expiration, namely [8]

$$\begin{aligned} C &= \rho^{-1}\left[pC_u + (1-p)C_d\right] \\ &= \rho^{-1}\left[p\left[\rho^{-1}\left[pC_{uu} + (1-p)C_{ud}\right]\right] + (1-p)\left[\rho^{-1}\left[pC_{ud} + (1-p)C_{dd}\right]\right]\right] \\ &= \rho^{-2}\left[p^2C_{uu} + 2p(1-p)C_{ud} + (1-p)^2C_{dd}\right]. \end{aligned}$$

It is also left to the reader to check that applying this formula with the paradigm created in the binary tree in Figure 1, we obtain that $C = 11.04$, agreeing with the

[8] Q3.4

previous result. To match the portfolio to the call price, we simply followed a predicted path in Figure 1 and adjusted the portfolio accordingly. This can always be done as long as the prediction is correct, as we are in a very simplified scenario (the stock price goes up or down at constant rates). In general, if the share price goes up, borrow more and get more shares. If the share price goes down sell stocks and pay off some debt. [9]

For practical matters, we want to see how we could make a profit if two periods before expiration, the call option can be bought or sold on the market for the price of e.g. 12.04, that is, the price of 1 more of the price 11.04 that what we found previously. We could follow the next strategy to achieve a profit of 1 without risk. Two periods before expiration, buy the matching portfolio for $S\Delta + B = 11.04$. By assumption, you can sell the call option for 1 more than the portfolio, namely 12.04. This results in a profit of exactly 1. Since we sold a call option we are still obliged to deliver on this. We do so in the next trading period. The matching portfolio will now by construction be worth exactly as much as the call. Simply sell the portfolio and buy the call to settle the debt. The net profit is exactly 1. Thus we can see that the arbitrage principle is violated and this example comes to reinforce our idea that $C = S\Delta + B$.

Now, we go back to the theoretical part as we want to generalize our previously obtained Equation 2 and consider n trading periods. During these periods, given the share price increases j times and decreases $n-j$ times, then the share price at expiration will be equal to $Su^j d^{n-j}$. We therefore have that $C_{u^j d^{n-j}} = \max\{0, u^j d^{n-j} S - K\}$, where we denoted the price of the corresponding call option at expiration by $C_{u^j d^{nj}}$. Since we want to find such a general equation for C depending on n, we work back from the expiration date and repeatedly apply our previously obtained $C = \rho^{-1}\left[pC_u + (1-p)C_d\right]$. We make an educated claim that [10]

$$C = \rho^{-n}\left(\sum_{j=0}^{n}\binom{n}{j}p^j(1-p)^{n-j}C_{u^j d^{n-j}}\right) \tag{3}$$

We leave the proof of this claim in the Appendix A.2. Deepening into the theoretical part, we now define the letter a to be the least positive integer such that $u^a d^{n-a} S > K$. We are interested in slightly changing Equation 3 so that we get rid of the dependency of $C_{u^j d^{n-j}}$, and shift to the dependency of S and K instead to have a better understanding on the effect of changes in the share and strike price on the call option price. [11] Recall that $C_{u^j d^{n-j}} = \max\{0, u^j d^{n-j} S - K\}$. Thus we can substitute this $C_{u^j d^{n-j}}$ expression into

[9] Q3.6
[10] Q3.7
[11] Q3.8

Equation 3 to get

$$C = \rho^{-n}\left(\sum_{j=0}^{n}\binom{n}{j}p^j(1-p)^{n-j}\max\{0, u^j d^{n-j}S - K\}\right).$$

Since a is defined as the the least positive integer such that $u^a d^{n-a}S > K$, we observe

$$C = \rho^{-n}\sum_{j=0}^{n}\binom{n}{j}p^j(1-p)^{n-j}\max\{0, u^j d^{n-j}S - K\} = \rho^{-n}\sum_{j=a}^{n}\binom{n}{j}p^j(1-p)^{n-j}(u^j d^{n-j}S - K).$$

as the terms

$$\sum_{j=0}^{a}\binom{n}{j}p^j(1-p)^{n-j}\max\{0, u^j d^{n-j}S - K\} = 0$$

due to $u^a d^{n-a}S \leq K$. Thus we conclude this section with

$$C = S\left(\sum_{j=a}^{n}\binom{n}{j}p^j(1-p)^{n-j}\frac{u^j d^{n-j}}{\rho^n}\right) - K\rho^{-n}\left(\sum_{j=a}^{n}\binom{n}{j}p^j(1-p)^{n-j}\right), \tag{4}$$

a consistent expression for the call option price C which depends on S, K, and ρ, and u and d as well as n. In the next section, we will introduce stochasticity into this formula and study the limit of this binomial model as $n \to \infty$ to infer the well-known Black–Scholes formula to price options.

4 The Black–Scholes Formula

In order to make our model more accurate, we must change the underlying assumptions. Firstly the binomial model only considers fluctuations between trading periods, while in actuality trading is continuous throughout the whole day. Furthermore the share can vary an infinite amount, it is severely reductive to assume it either increases or decreases at the same rate. In this section we will therefore elaborate and build out the model to more accurately fit real life.

The first thing we can do is shorten the trading times to minutes or even seconds, but of course the fluctuation in the share price has to also change. After all the share can change much more in a week than in a minute. Likewise the values u, d and q will also depend on the length of the trading period. Naturally, the shorter the trading period, the closer both u and d will be to 1.

In order to mathematically describe this, let t denote the time (in years) until the option's expiration, and divide t into n time intervals of equal length and call this length

h, so $h = t/n$. The remaining term t itself will not depend on the number of trading periods. The more often trading takes place, the smaller h. We now must express u, d and ρ in terms of n in such a way that their limits as $n \to \infty$ match more realistic results. Let us start with ρ. Let $\hat{r}$ be the interest rate over one trading period, meaning $\rho = 1+\hat{r}$, and let r be the risk-free interest rate. The interest rate ρ^n over n trading periods over a duration of t years must then be equal to e^{rt}, that is, $\rho = \rho_n = e^{rt/n}$.

We now determine u_n and d_n. Recall u to be the coefficient of increase of the share as it goes up and d to be the coefficient of decrease of the share as it goes down. As we take infinitesimal trading periods, less trading is possible, meaning less variation and so logically both u_n and d_n have to be asymptotically closer and closer to 1. A practical way to realize this is using the annual volatility σ and letting u_n and d_n be each other's reciprocals. A convenient choice is therefore

$$u_n = e^{\sigma\sqrt{t/n}}\ , \qquad d_n = e^{-\sigma\sqrt{t/n}}\ .$$

for some $\sigma > 0$. Note that $u_n \cdot d_n = 1$, meaning the increase is the same as the decrease over one trading period. We can now determine p_n, since it depends entirely on the previously determined parameters. Plugging those in yields

$$p_n = \frac{\rho_n - d_n}{u_n - d_n} = \frac{e^{rt/n} - e^{-\sigma\sqrt{t/n}}}{e^{\sigma\sqrt{t/n}} - e^{-\sigma\sqrt{t/n}}} = \frac{e^{rt/n+\sigma\sqrt{t/n}} - 1}{e^{2\sigma\sqrt{t/n}} - 1}\ .$$

The final parameter to determine is a_n (now dependent on n). Recall that a is the smallest integer such that $u^a d^{n-a} S > K$ by construction. We plug in this inequality our new definitions for u_n and d_n and solve it for a_n to obtain the inequality

$$a_n > \frac{\log(K/S) - n\log d_n}{\log(u_n/d_n)} =: A.$$

By definition a_n is the smallest integer above this bound. This naturally implies the existence of an $\varepsilon_n \in [0, 1)$ for all n such that [12]

$$a_n - 1 = \frac{\log(K/S) - n\log d_n}{\log(u_n/d_n)} - \varepsilon_n. \tag{5}$$

We will make use of this equality to rewrite the last sum of Equation 4. The probability theory versed reader may recognize this last sum term to be the cumulative distribution function of a binomial random variable. We will use this observation to rewrite this sum

[12] Q4.1

with a binomial random variable with parameters n and p_n denoted by $N_n \sim Bin(n, p_n)$. Notice that

$$\sum_{j=a_n}^{n} \binom{n}{j} p_n^j (1-p_n)^{n-j} = P(N_n \geq a_n) = 1 - P(N_n \leq a_n - 1). \tag{6}$$

The motivation underneath this recontextualization is that we want to approximate $P(N_n \leq a_n - 1)$ by the standard normal distribution using the theory behind *Theorem 4.1* and the Central Limit Theorem (CLT). Note that in our case, p is not fixed but varies with n. The CLT still holds provided that $p = p_n$ converges to a constant in (0, 1), and this will be the case as we will see later on when we study the limit of p_n. Since we want to reach some conclusions as $n \to \infty$, we first make the following claim [13]

Theorem 4.1. *If for all $y \in \mathbb{R}$, we have $\mathbb{P}(Y_n \leq y) \to \Phi(y)$ and if $x_n \to x$ for some $x \in \mathbb{R}$ as $n \to \infty$, then we also have $\mathbb{P}(Y_n \leq x_n) \to \Phi(x)$.*

The proof of this theorem can be found in the Appendix section. See A.3 proof. Now that we set up the theoretical framework, we will rewrite the last probability from Equation 6 using our bound for a_n from Equation 5 to obtain

$$\mathbb{P}(N_n \leq a_n - 1) = \mathbb{P}\left(N_n \leq A\right) = \mathbb{P}\left(\frac{N_n - np_n}{\sqrt{np_n(1-p_n)}} \leq \frac{A - np_n}{\sqrt{np_n(1-p_n)}}\right)$$

In our case, the first hypothesis $\mathbb{P}(Y_n \leq y) \to \Phi(y)\ \forall y$ of *Theorem 4.1* holds by the CLT, and it is going to be seen $x_n = \frac{A - np_n}{\sqrt{np_n(1-p_n)}}$ converges to some $x \in \mathbb{R}$. Therefore, this theorem enables us to make the standard normal distribution approximation $\mathbb{P}(Y_n \leq x_n) \to \Phi(x)$ for $\mathbb{P}(N_n \leq a_n - 1)$. Notice from the first equality that, since we are studying a discrete probability distribution, we can use A instead of $a_n - 1$, as $a_n - 1 = A - \varepsilon_n$. By applying a first step of the aforementioned theorem (we still need to study the limit of the term inside Φ), we see that as $n \to \infty$ we have

$$\mathbb{P}(N_n \leq a_n - 1) \xrightarrow{\text{d}} \Phi\left(\frac{\frac{\log(K/S) - n\log d_n}{\log(u_n/d_n)} - np_n}{\sqrt{np_n(1-p_n)}}\right) = \Phi\left(\frac{\frac{\log(K/S)}{2\sigma\sqrt{t/n}} - n\left(p_n - \frac{1}{2}\right)}{\sqrt{np_n(1-p_n)}}\right)$$

For the equality step, use that $u_n d_n = 1$ and therefore $\log(u_n/d_n) = -2\log d_n$. Also, for the denominator of the remaining log term, use the definitions of $u_n = e^{\sigma\sqrt{t/n}}$ and $d_n = e^{-\sigma\sqrt{t/n}}$ to see that $\log(u_n/d_n) = 2\sigma\sqrt{t/n}$. Lastly, factor out n in $n/2 - np_n$ and

[13] Q4.2

divide both numerator and denominator of the whole fraction by $\sqrt{n}$ to obtain

$$\mathbb{P}(N_n \leq a_n - 1) \to \Phi\left(\frac{\log(K/S)}{2\sigma\sqrt{t}\sqrt{p_n(1-p_n)}} - \frac{\sqrt{n}\left(p_n - \frac{1}{2}\right)}{\sqrt{p_n(1-p_n)}}\right).$$

For practical matters, we claim the following two limits to be true, see Appendix A.3 for the proof

$$\lim_{n\to\infty} p_n = \frac{1}{2} \quad \text{and} \quad \lim_{n\to\infty} \sqrt{n}\left(p_n - \frac{1}{2}\right) = \frac{rt - \frac{1}{2}\sigma^2 t}{2\sigma\sqrt{t}}\ . \tag{7}$$

Using these two limits and the fact that the limit of a product is the product of the limit we get $\lim_{n\to\infty}\sqrt{p_n(1-p_n)} = \frac{1}{2}$, it is straightforward to see that as $n \to \infty$ we have

$$\Phi\left(\frac{\log(K/S)}{2\sigma\sqrt{t}\sqrt{p_n(1-p_n)}} - \frac{\sqrt{n}\left(p_n - \frac{1}{2}\right)}{\sqrt{p_n(1-p_n)}}\right) \to \Phi\left(\frac{\log(K/S)}{\sigma\sqrt{t}} + \frac{-rt + \frac{1}{2}\sigma^2 t}{\sigma\sqrt{t}}\right).$$

From this, use $-rt = \log(e^{-rt})$, an essential sum property of the logarithms and simplify some terms to finally see that

$$\Phi\left(\frac{\log(K/S)}{\sigma\sqrt{t}} + \frac{-rt + \frac{1}{2}\sigma^2 t}{\sigma\sqrt{t}}\right) = \Phi\left(\frac{\log(Ke^{-rt}/S)}{\sigma\sqrt{t}} + \frac{1}{2}\sigma\sqrt{t}\right).$$

Therefore, we got a suitable approximation for $\mathbb{P}(N_n \leq a_n - 1)$ and by means of this expression and all this previous reasoning we infer a stochastic approximation for the last sum of Equation 4, namely

$$\sum_{j=0}^{n}\binom{n}{j}p_n^j(1-p_n)^{n-j} = 1 - \mathbb{P}(N_n \leq a_n - 1) \to 1 - \Phi\left(\frac{\log(Ke^{-rt}/S)}{\sigma\sqrt{t}} + \frac{1}{2}\sigma\sqrt{t}\right).$$

Note that we could also derive an approximation for the first sum of Equation 4 which would give us a final formula to price call options with asymptotic n, that is, with continuous trading. This aforementioned formula will be known as Black–Scholes formula for call options. So, we do an analogous process for the first sum. For this purpose, let us introduce a new term $p'_n = (u_n/\rho_n)p_n$ for which it holds that $1 - p'_n = (d_n/\rho_n)(1-p_n)$. See Appendix A.3 for the proof of this claim.

Using the new term p'_n and the claim, we observe for the first sum term that

$$\sum_{j=a_n}^{n}\binom{n}{j}p_n^j(1-p_n)^{n-j}\frac{u_n^j d_n^{n-j}}{\rho_n^n} = \sum_{j=a_n}^{n}\binom{n}{j}{p'_n}^j(1-p'_n)^{n-j} = 1 - \mathbb{P}(N'_n \leq a_n - 1)$$

where we created a new random variable $N'_n \sim Bin(n, p'_n)$. From here, we use the same reasoning as we did to get the approximation for the last sum term. We now have

$$\mathbb{P}(N'_n \leq a_n - 1) = \mathbb{P}\left(N'_n \leq A\right) = \mathbb{P}\left(\frac{N'_n - np'_n}{\sqrt{np'_n(1-p'_n)}} \leq \frac{A - np'_n}{\sqrt{np'_n(1-p'_n)}}\right)$$

By applying again *Theorem 4.1* we see that as $n \to \infty$ we have

$$\mathbb{P}(N'_n \leq a_n - 1) \to \Phi\left(\frac{\log(K/S)}{2\sigma\sqrt{t}\sqrt{p'_n(1-p'_n)}} - \frac{\sqrt{n}\left(p'_n - \frac{1}{2}\right)}{\sqrt{p'_n(1-p'_n)}}\right).$$

Now, we again make new claims about two more limits of special interest with their proofs in Appendix A.3, namely that [14]

$$\lim_{n\to\infty} p'_n = \frac{1}{2} \quad \text{and} \quad \lim_{n\to\infty} \sqrt{n}\left(p'_n - \frac{1}{2}\right) = \frac{rt + \frac{1}{2}\sigma^2 t}{2\sigma\sqrt{t}},$$

and using the same logarithm properties and limit theorem simplification method as above, we therefore conclude that

$$\sum_{j=a_n}^{n} \binom{n}{j} p_n^j (1-p_n)^{n-j} \frac{u_n^j d_n^{n-j}}{\rho_n^n} = 1 - \mathbb{P}(N'_n \leq a_n - 1) \to 1 - \Phi\left(\frac{\log(Ke^{-rt}/S)}{\sigma\sqrt{t}} - \frac{1}{2}\sigma\sqrt{t}\right).$$

Now [15] that we have an asymptotic expression for both sums in equation (4), all left to do is replacing these expressions for the asymptotic sums into the aforementioned equation. Doing this, we finally obtain the Black–Scholes formula for call options, i.e.

$$\begin{aligned}
C &= S\left(\sum_{j=a_n}^{n} \binom{n}{j} p_n^j (1-p_n)^{n-j} \frac{u_n^j d_n^{n-j}}{\rho_n^n}\right) - K\rho_n^{-n}\left(\sum_{j=a_n}^{n} \binom{n}{j} p_n^j (1-p_n)^{n-j}\right) \\
&\to S\left(1 - \Phi\left(\frac{\log((Ke^{-rt})/S)}{\sigma\sqrt{t}} - \frac{1}{2}\sigma\sqrt{t}\right)\right) - Ke^{(rt/n)-n}\left(1 - \Phi\left(\frac{\log((Ke^{-rt})/S)}{\sigma\sqrt{t}} + \frac{1}{2}\sigma\sqrt{t}\right)\right) \\
&= S\Phi\left(\frac{\log(S/(Ke^{-rt}))}{\sigma\sqrt{t}} + \frac{1}{2}\sigma\sqrt{t}\right) - Ke^{-rt}\Phi\left(\frac{\log(S/(Ke^{-rt}))}{\sigma\sqrt{t}} - \frac{1}{2}\sigma\sqrt{t}\right), \qquad (8)
\end{aligned}$$

where the last step is a result of the probability theory law $\Phi(z) = 1 - \Phi(-z)$. Having the price of a call option, the price of a put option is now no longer difficult to determine, and it follows from the so-called put-call parity for European options (which we will not prove)

$$P = C - S + Ke^{-rt}\ .$$

[14] Q4.3
[15] Q4.4

Combining this together with Equation 8, we obtain Black–Scholes formula for put options [16]

$$
\begin{aligned}
P &= C - S + Ke^{-rt} \\
&= S\left(\Phi\left(\frac{\log(S/(Ke^{-rt}))}{\sigma\sqrt{t}} + \frac{1}{2}\sigma\sqrt{t}\right) - 1\right) + Ke^{-rt}\left(1 - \Phi\left(\frac{\log(S/(Ke^{-rt}))}{\sigma\sqrt{t}} - \frac{1}{2}\sigma\sqrt{t}\right)\right) \\
&= -S\Phi\left(\frac{\log((Ke^{-rt})/S)}{\sigma\sqrt{t}} - \frac{1}{2}\sigma\sqrt{t}\right) + Ke^{-rt}\Phi\left(\frac{\log((Ke^{-rt})/S)}{\sigma\sqrt{t}} + \frac{1}{2}\sigma\sqrt{t}\right)
\end{aligned}
$$

We close this section at this result. To conclude the paper we will apply the Black–Scholes for real world data.

5 Applying the Black–Scholes formula

In order to apply the Black–Scholes formula we need values for the variables S, K, t, σ and r. Data is readily available for both stock prices S and the interest rate r. Once a specific option is chosen, both K and t will be fixed as well. Since t is expressed in years, we can approximate it by taking the number of trading days left until expiration and dividing by 250. If the option's term is not too short (more than a month), one can also use the number of calendar days until the expiration date divided by 365 as an approximation. The only thing left to determine is the annual volatility σ. To illustrate how to estimate this, we will use the weekly closing prices of a Hagemeyer share in the year 1996 as an example, see Table 2.

Date	Closing price	Date	Closing price
19/04/96	113.20	14/06/96	121.20
26/04/96	116.70	21/06/96	121.00
03/05/96	115.50	28/06/96	121.60
10/05/96	118.00	05/07/96	130.00
17/05/96	115.30	12/07/96	129.00
24/05/96	116.60	19/07/96	131.00
31/05/96	117.50	26/07/96	129.00
07/06/96	124.80	02/08/96	124.00

Table 2: Weekly closing prices of a Hagemeyer share in 1996

In contrast to more modern models, the main assumption of the Black–Scholes model

[16] Q4.5

is assuming that the logarithm of the relative price, being the current price divided by the previous price, over a fixed time period is normally distributed with expected value and variance proportional to the period length. However, when the Black–Scholes was put to practice, it became evident that this assumption was wrong as the tails of the normal distribution flattened too quickly to match reality. In this case, it is moreover assumed that the relative prices are independent of one another.

We circle back to the Hagemeyer shares to illustrate how to put this to practice in order to estimate the annual volatility. The first four prices in Table 2 are 113.20, 116.70, 115.50, and 118.00, so the first three relative prices are $116.70/113.20 = 1.031$, $115.30/116.70 = 0.990$, and $118.00/115.50 = 1.022$. The natural logarithms of these prices are $\log(1.031) = 0.0305$, $\log(0.990) = -0.0101$, and $\log(1.022) = 0.0218$. Under the conditions mentioned above, these three form an independent sample from a normal distribution with standard deviation $\tilde{\sigma}$, the weekly volatility of the share. If we use all the data in Table 2, we have an independent sample of length 15 of this normal distribution. In general, if S_k is the share price in week k, then it is assumed that $\log(R_k) = \log(S_k/S_{k-1})$ is normally distributed with parameters $\tilde{\mu}$ and $\tilde{\sigma}^2$. An unbiased estimator for $\tilde{\sigma}^2$ is then given by [17]

$$\widehat{\tilde{\sigma}^2} = \frac{1}{n-1}\sum_{k=1}^{n}(\log R_k - \hat{\tilde{\mu}})^2 \qquad \text{with} \qquad \hat{\tilde{\mu}} = \frac{1}{n}\sum_{k=1}^{n}\log R_k \ .$$

Finally, to estimate the annual volatility for the Hagemeyer shares we take the square root of $52 \cdot \widehat{\tilde{\sigma}^2}$, where we multiply by 52 as we have taken week intervals in our data. Recall that $\sqrt{\widehat{\tilde{\sigma}^2}}$ approximates the *weekly* volatility. Doing these calculations gives us $\hat{\tilde{\mu}} = 0.0061$ and $\widehat{\tilde{\sigma}^2} = 0.0009$, resulting in an annual volatility of 0.2175.

With this example out of the way, we could finally turn to current real world data in order to compare our prices with those on the current options exchange, as this model is but a stepping stone for modern day pricing techniques. However, we could not find data that fit our chosen stock so we decided to pursue an empirical study with artificial data, namely, computed prices by means of the Black–Scholes formula. Table 3 shows the 40 most recent days of closing prices for the Ahold Delhaize share (as of writing). Using the same method as above, we obtain $\hat{\tilde{\mu}} = 0.0013$ and $\widehat{\tilde{\sigma}^2} = 0.0001$. To get the annual volatility, we must multiply by 250 instead of 52, as we have taken a sample of days instead of weeks. This results in an annual volatility of $\sigma = 0.17129$. [18]

The only parameter left to determine in order to apply Black–Scholes in is the so-

[17] Q5.1
[18] Q5.2

Date	Close	Date	Close	Date	Close	Date	Close
22/11/22	28.75	07/12/22	29.73	21/12/22	29.35	06/01/23	29.05
23/11/22	28.91	08/12/22	29.91	22/12/22	29.44	09/01/23	29.60
25/11/22	29.17	09/12/22	29.95	23/12/22	29.46	10/01/23	29.35
28/11/22	28.88	12/12/22	29.68	27/12/22	29.34	11/01/23	29.94
29/11/22	28.94	13/12/22	29.76	28/12/22	28.91	12/01/23	30.25
30/11/22	29.09	14/12/22	29.39	29/12/22	29.09	13/01/23	30.22
01/12/22	29.60	15/12/22	29.07	30/12/22	28.70	17/01/23	30.74
02/12/22	29.60	16/12/22	28.75	03/01/23	28.86	18/01/23	29.96
05/12/22	29.25	19/12/22	28.82	04/01/23	29.24	19/01/23	30.11
06/12/22	29.55	20/12/22	29.15	05/01/23	28.75	20/01/23	30.27

Table 3: Closing share price for the past 40 days of the Ahold stock (as of writing).

called risk-free interest rate r parameter. However, we encounter a problem which is that we considered a risk-free interest rate r, which in reality does not exist, therefore we need an estimator for such. For calculations with the Black–Scholes formula, within the Eurozone, the so-called euribor-interest rate (Euro Interbank Offered Rate) is often used. This euribor-interest rate is roughly speaking the average interest rate of mutual loans between banks within the Eurozone with very good credit ratings, and we consider it as a suitable estimate for r given some value of t. We calculated both the call and put option prices for 1, 2, 3, 24 and 36 months using the derived Black–Scholes formula. The results[19] are displayed in Table 4, with t being expressed in trading days.

	$K \downarrow \backslash t \rightarrow$	20	40	60	500	750
Call	29	1.4594	1.6764	1.8695	4.8262	6.6398
	32	0.1011	0.2755	0.4439	3.2484	5.0067
Put	29	0.1386	0.2997	0.4318	1.3529	1.2280
	32	1.7751	1.8878	1.9888	2.5471	2.1664

Table 4: Results of the Black–Scholes formula applied to Ahold options.

We took $S = 30.27$, the latest data point available in Table 3, and used the previously calculated annual volatility of $\sigma = 0.17129$. We choose strike prices both above and below the share price to illustrate slight differences.

For the sake of illustration, let‘s take a look at the behaviour of those options with terms 24 and 36 months. We want to see how the behaviour of the share price S affects

[19] Q5.3

these options prices. Suppose a year later that these options have been priced as in Table 4, the share price increases by 5, then what is the new price of these call options? Note that the only thing we have to do is using again the Black–Scholes formula with updates in t (that is, previous t minus 250 trading days), the updated share price S and new interest rates r as they are related with t as well. Doing so we obtain the following results.

	$K \downarrow \backslash t \rightarrow$	250	500
Call	29	7.4930	8.9484
	32	5.0868	6.7628
Put	29	0.2205	0.4751
	32	0.7106	1.0615

Table 5: Updated prices of Ahold options after one year and an increase of 5 in S.

Recall that these two columns represent the same four options as those from the two rightmost columns in Table 4. However, some external conditions have changed, namely there is one year less in the options term and the share price S has increased by 5. Results in Table 5 represent how their prices changed under these new circumstances. As we expected, following from conceptual conclusions that we had already summarized previously in Table 1, the call options prices increase with the increase in the share price S and the opposite effect happened for the put options, which makes intuitive sense. If the share goes from 30.27 to 35.27, buying the right to buy the share (in 1-2 year period) at the same strike price must be more expensive. Conceptually, it would seem to be more likely that I could use my right, and therefore the expected value of the the call option would be bigger. Reasonably, we could expect an adversary effect if the share price S goes down instead.

To end this report, we want to illustrate the difference in growth of the share price and the call option price expiring in 3 years. For that purpose, we will look at the correlation between the share price and the option price for a given example. In order to accurately do this we sampled the weekly closing prices of the Ahold share over the past year and estimated the option prices according to the Black–Scholes model for each of the 52 entries. We took strike prices by rounding down the closing prices, we used the same σ as before and kept r as the euribor estimate for the proper term. In order to compare growth between share and call, we get an increase rate from one week to the next by dividing the current by the previous price for both the share and the call. In Table 6, we present all this data gathered.

Date	Close	Percentual Increase	Call Price	Percentual Increase
24/01/22	32.3300	1.0158	7.7285	1.0501
31/01/22	32.8400	1.0033	8.1160	1.0239
07/02/22	32.9500	0.9308	8.3099	0.9115
14/02/22	30.6700	0.9896	7.5747	0.9609
21/02/22	30.3500	0.9891	7.2788	0.9587
28/02/22	30.0200	0.9930	6.9781	1.0731
07/03/22	29.8100	1.0480	7.4882	0.9955
14/03/22	31.2400	1.0080	7.4543	1.0232
21/03/22	31.4900	1.0187	7.6270	0.9822
28/03/22	32.0800	1.0349	7.4909	1.0625
04/04/22	33.2000	0.9220	7.9588	0.9495
11/04/22	30.6100	0.9846	7.5566	0.9447
18/04/22	30.1400	0.9754	7.1385	0.9928
25/04/22	29.4000	0.9830	7.0869	1.0383
02/05/22	28.9000	0.9661	7.3584	0.9663
09/05/22	27.9200	0.9473	7.1108	0.9080
16/05/22	26.4500	1.0476	6.4565	1.0652
23/05/22	27.7100	0.9960	6.8777	0.9825
30/05/22	27.6000	0.9714	6.7576	1.0039
06/06/22	26.8100	0.9847	6.7837	0.9457
13/06/22	26.4000	0.9996	6.4152	0.9944
20/06/22	26.3900	0.9981	6.3796	0.9893
27/06/22	26.3400	1.0224	6.3110	1.0952
04/07/22	26.9300	0.9747	6.9121	0.9142
11/07/22	26.2500	1.0282	6.3192	1.0931
18/07/22	26.9900	1.0219	6.9073	0.9801
25/07/22	27.5800	0.9793	6.7696	0.9434
01/08/22	27.0100	1.0322	6.3865	1.1082
08/08/22	27.8800	1.0201	7.0772	0.9773
15/08/22	28.4400	0.9641	6.9163	0.9597
22/08/22	27.4200	0.9916	6.6373	0.9673
29/08/22	27.1900	1.0405	6.4202	1.0672
05/09/22	28.2900	0.9788	6.8516	1.0097
12/09/22	27.6900	0.9372	6.9182	0.9598
19/09/22	25.9500	0.9803	6.6401	0.9319
26/09/22	25.4400	0.9780	6.1878	1.0341
03/10/22	24.8800	1.0707	6.3986	1.0380
10/10/22	26.6400	1.0154	6.6416	0.9580
17/10/22	27.0500	1.0359	6.3627	1.0284
24/10/22	28.0200	1.0011	6.5433	0.9992
31/10/22	28.0500	1.0089	6.5383	1.0447
07/11/22	28.3000	1.0046	6.8305	1.0114
14/11/22	28.4300	1.0260	6.9081	0.9986
21/11/22	29.1700	1.0147	6.8984	1.0468
28/11/22	29.6000	1.0118	7.2213	1.0575
05/12/22	29.9500	0.9599	7.6362	0.9433
12/12/22	28.7500	1.0247	7.2032	0.9950
19/12/22	29.4600	0.9742	7.1669	0.9910
26/12/22	28.7000	1.0122	7.1022	0.9703
02/01/23	29.0500	1.0403	6.8915	1.0499
09/01/23	30.2200	1.0017	7.2352	1.0013
16/01/23	30.2700	0.9742	7.2447	0.9884
23/01/23	29.4900		7.1611	0.0000

Table 6: Weekly data for Ahold share prices and the B–S calculated call options prices.

We performed a small bivariate analysis on the variables that measured the percentual increases of both the shares and the call options, namely we looked at the correlation of these variables which turned out to be 0.65 [20]. This means that indeed, as we already expected, both variables are correlated. As a proxy, it seems we could explain 65% of the behaviour of the assets we set up (Ahold call options with the given characteristics) by looking at the behaviour of the share prices. We conclude the report with this small empirical result that supports a previous theoretical result we derived earlier, namely, the correlation exposed between S and C in Table 1.

[20] See the dataset here

A Proofs

A.1 The price of an option

Theorem A.1. *The sequence* $\left(1+\frac{r}{n}\right)^n$ *is increasing with respect to* n*, where* r *is a fixed real number.*

<u>Proof:</u> We could start by looking at the derivative but we realised calculations derived into difficult terms. We look at the division of two arbitrary consecutive terms and we want to prove this is bigger than 1. Observe that we want to prove

$$\frac{\left(1+\frac{r}{n+1}\right)^{n+1}}{\left(1+\frac{r}{n}\right)^n} = \left(\frac{1+\frac{r}{n+1}}{1+\frac{r}{n}}\right)^n \left(1+\frac{r}{n+1}\right) = \left(\frac{n^2+n(r+1)}{n^2+n(r+1)+r}\right)^n \left(1+\frac{r}{n+1}\right) > 1,$$

and this is equivalent to

$$\left(1-\frac{r}{n^2+n(r+1)+r}\right)^n \left(1+\frac{r}{n+1}\right) = \left(1-\frac{nr}{n^2+n(r+1)+r}\right)\left(1+\frac{r}{n+1}\right) > 1,$$

since $(1-x)^n \geq 1-nx$ by the Bernoulli inequality. All this equates to

$$\left(\frac{n^2+n+r}{n^2+n(r+1)+r}\right)\left(\frac{n+(r+1)}{n+1}\right) > 1$$

which is equivalent to

$$(n^2+n+r)(n+r+1) - (n^2+n(r+1)+r)(n+1) > 0.$$

Note that the last expression we obtained simplifies to

$$n^3+n^2(r+2)+n(2r+1)+r(r+1) - (n^3+n^2(r+2)+n(2r+1)+r) = r^2 > 0.$$

Since the last expression is clearly true and we just followed a chain of equivalences, then the first expression is also true and we proved the claim. ■

Go back to the text clicking here 2.1

Theorem A.2. *The limit of the function* $(1+\frac{r}{n})^n$ *as* $n \to \infty$ *is equal to* e^r*.*

<u>Proof:</u> Firstly, we use the fact that $x = e^{ln(x)}$, to say then that

$$\lim_{n\to\infty}(1+\frac{r}{n})^n = \lim_{n\to\infty}\exp(ln((1+\frac{r}{n})^n)) = \lim_{n\to\infty}\exp(n\cdot ln(1+\frac{r}{n})) = \lim_{n\to\infty}\exp\left(\frac{ln(1+\frac{r}{n})}{\frac{1}{n}}\right).$$

Since we know that

$$\lim_{n\to\infty} \exp(n) = \exp(\lim_{n\to\infty} n),$$

then we can pick up where we left off and say that

$$\lim_{n\to\infty} \exp\left(\frac{ln(1+\frac{r}{n})}{\frac{1}{n}}\right) = \exp\left(\lim_{n\to\infty} \frac{ln(1+\frac{r}{n})}{\frac{1}{n}}\right).$$

We are now in a situation where if we evaluate the limit of $n \to \infty$, the result is an indeterminate form of $\frac{ln(1+0)}{0} = \frac{0}{0}$, thus we know we can apply L'Hôpital's formula to manipulate the argument of the limit. Therefore we have that

$$\exp\left(\lim_{n\to\infty} \frac{ln(1+\frac{r}{n})}{\frac{1}{n}}\right) = \exp\left(\lim_{n\to\infty} \frac{\frac{-r}{n^2}/1+\frac{r}{n}}{\frac{-1}{n^2}}\right) = \exp\left(\lim_{n\to\infty} \frac{r}{1+\frac{r}{n}}\right).$$

Now, we are finally able to evaluate the limit and result in a determinate form, where we can see that

$$\lim_{n\to\infty} (1+\frac{r}{n})^n = \exp\left(\lim_{n\to\infty} \frac{r}{1+\frac{r}{n}}\right) = \exp\left(\frac{r}{1+0}\right) = \exp(r) = e^r.$$

■

Go back to the text clicking here 2.2

A.2 A binomial model for option prices

<u>Claim:</u> If we choose Δ and B to be as as solved in Equation 1, then one period before expiration the value of our portfolio will also be equal to the price of the call option, that is, $C = S\Delta + B$.

<u>Proof:</u>

This claim is proved by again using the arbitrage principle, this time in two steps: $C > S\Delta + B$, and $C < S\Delta + B$. It is important to recall that the share price can rise or drop. However, we chose Δ and B so that we can ensure that in both cases the value of the portfolio after this one trading period is equal to the price of the call option.

Suppose $C > S\Delta + B$. Then one can always achieve a profit with the following strategy. Sell the call option at the price C, and buy the replicating portfolio (i.e. buy Δ amount of shares at price S each and borrow the amount B). This yields a net amount of $C - (S\Delta + B) > 0$ euro. In the next trading period, make a simple neutral transaction, i.e. sell the portfolio and use this money to buy the call option (remember their prices

will match as the choices of Δ and B guarantee this property). The net balance is thus $C - (S\Delta + B) > 0$ euro.

Suppose $C < S\Delta + B$. Then one can also always achieve a profit with the following strategy. Buy the call option at the price C, and sell the replicating portfolio (i.e. sell Δ amount of shares at price S each and deposit the amount B in an interest rate account). Note that for this second strategy we need an extra assumption, that is, we already owned the portfolio. This yields a net amount of $S\Delta + B - C > 0$ euro. In the next trading period, make again another simple neutral transaction, i.e. sell the call option and use this money to buy the replicating portfolio (their prices will also match as the choices of Δ and B guarantee this property). Your net balance is thus $S\Delta + B - C > 0$ euro.

Following from the arbitrage principle, we conclude that $C = S\Delta + B$ is the right price for the call option. ■

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<u>Claim:</u> We make an educated claim that the price of a call option is generalized so that

$$C = \rho^{-n}\left(\sum_{j=0}^{n}\binom{n}{j}p^j(1-p)^{n-j}C_{u^j d^{n-j}}\right) \tag{9}$$

<u>Proof:</u> We will prove this identity by induction. The basis case $n = 1$ is true as we have $C = \rho^{-1}[pC_u + (1-p)C_d]$ which is known to be true as we previously got to that conclusion. Assume the induction hypothesis (9) holds for all integer up to n. Applying the induction step,

$$\begin{aligned}
C &= \rho^{-n}\left(\sum_{j=0}^{n}\binom{n}{j}p^j(1-p)^{n-j}C_{u^j d^{n-j}}\right),\\
&= \rho^{-n}\left(\sum_{j=0}^{n}\binom{n}{j}p^j(1-p)^{n-j}\rho^{-1}\left[pC_{u^{j+1}d^{n-j}} + (1-p)C_{u^j d^{n-j+1}}\right]\right),\\
&= \rho^{-(n+1)}\left(\sum_{j=0}^{n}\binom{n}{j}p^{j+1}(1-p)^{n-j}C_{u^{j+1}d^{n-j}} + p^j(1-p)^{n-j+1}C_{u^j d^{n-j+1}}\right).
\end{aligned}$$

We continue by distributing the summation and re-indexing the first term

$$C = \rho^{-(n+1)}\left(\sum_{j=1}^{n+1}\binom{n}{j-1}p^j(1-p)^{n-j+1}C_{u^j d^{n-j+1}} + \sum_{j=0}^{n}\binom{n}{j}p^j(1-p)^{n-j+1}C_{u^j d^{n-j+1}}\right).$$

Now we make use of the following known combinatorial identity

$$\binom{n}{j-1} + \binom{n}{j} = \binom{n+1}{j},$$

for the terms of the sum going from $j = 1$ to $j = n$ and compute $j = 0$ and $j = n + 1$ separately to obtain

$$\begin{aligned} C &= \rho^{-(n+1)} \left(\sum_{j=1}^{n} \binom{n+1}{j} p^j (1-p)^{n-j+1} C_{u^j d^{n-j+1}} + p^{n+1} C_{u^{n+1}} + (1-p)^{n+1} C_{d^{n+1}} \right) \\ &= \rho^{-(n+1)} \left(\sum_{j=0}^{n+1} \binom{n+1}{j} p^j (1-p)^{n-j+1} C_{u^j d^{n-j+1}} \right), \end{aligned}$$

finishing the proof by induction. Thus we have deduced a general equation for the price of a call option after n trading terms, depending on n, ρ, u, and d. ■

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A.3 The Black–Scholes formula

<u>Claim:</u> Let p_n be

$$p_n = \frac{\rho_n - d_n}{u_n - d_n} = \frac{e^{rt/n} - e^{-\sigma\sqrt{t/n}}}{e^{\sigma\sqrt{t/n}} - e^{-\sigma\sqrt{t/n}}} = \frac{e^{rt/n+\sigma\sqrt{t/n}} - 1}{e^{2\sigma\sqrt{t/n}} - 1} \ .$$

Then, we have

$$\lim_{n\to\infty} p_n = \frac{1}{2} \quad \text{and} \quad \lim_{n\to\infty} \sqrt{n}\left(p_n - \frac{1}{2}\right) = \frac{rt - \frac{1}{2}\sigma^2 t}{2\sigma\sqrt{t}} \ .$$

<u>Proof:</u> We make use of the Taylor expansion of the rightmost identity for p_n. Observe that

$$p_n = \frac{e^{rt/n+\sigma\sqrt{t/n}} - 1}{e^{2\sigma\sqrt{t/n}} - 1} = \frac{rt/n + \sigma\sqrt{t/n} + \frac{1}{2}\sigma^2 t/n + O(n^{-3/2})}{2\sigma\sqrt{t/n} + 2\sigma^2 t/n + O(n^{-3/2})}.$$

Multiplying both numerator and denominator by $\sqrt{n}$ and grouping the terms using their order of n, we get

$$p_n = \frac{\sigma\sqrt{t} + \frac{1}{\sqrt{n}}\left(rt + \frac{1}{2}\sigma^2 t\right) + O(n^{-1})}{2\sigma\sqrt{t} + 2\sigma^2 t/\sqrt{n} + O(n^{-1})} = \frac{\sigma\sqrt{t} + O(n^{-1/2})}{2\sigma\sqrt{t} + O(n^{-1/2})} \tag{10}$$

Thus taking the limit as $n \to \infty$ in this expression results in

$$\lim_{n\to\infty} p_n = \lim_{n\to\infty} \frac{\sigma\sqrt{t} + O(n^{-1/2})}{2\sigma\sqrt{t} + O(n^{-1/2})} = \frac{\sigma\sqrt{t}}{2\sigma\sqrt{t}} = \frac{1}{2}$$

For the second limit, we use the middle expression in (10) and put the minus term inside the numerator so that

$$\sqrt{n}\left(p_n - \frac{1}{2}\right) = \sqrt{n}\left(\frac{\sigma\sqrt{t} + \frac{1}{\sqrt{n}}\left(rt + \frac{1}{2}\sigma^2 t\right) + O(n^{-1})}{2\sigma\sqrt{t} + 2\sigma^2 t/\sqrt{n} + O(n^{-1})} - \frac{1}{2}\right)$$
$$= \sqrt{n}\left(\frac{\frac{1}{\sqrt{n}}\left(rt - \frac{1}{2}\sigma^2 t\right) + O(n^{-1})}{2\sigma\sqrt{t} + 2\sigma^2 t/\sqrt{n} + O(n^{-1})}\right).$$

Thus, note that the square roots of n cancel out and we obtain

$$\lim_{n\to\infty}\sqrt{n}\left(p_n - \frac{1}{2}\right) = \lim_{n\to\infty}\left(\frac{rt - \frac{1}{2}\sigma^2 t + O(n^{-1/2})}{2\sigma\sqrt{t} + O(n^{-1/2})}\right) = \frac{rt - \frac{1}{2}\sigma^2 t}{2\sigma\sqrt{t}}$$

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<u>Claim:</u> Let $p'_n = \frac{u_n}{\rho_n} p_n$. Then, we have

$$\lim_{n\to\infty} p'_n = \frac{1}{2} \quad \text{and} \quad \lim_{n\to\infty}\sqrt{n}\left(p'_n - \frac{1}{2}\right) = \frac{rt + \frac{1}{2}\sigma^2 t}{2\sigma\sqrt{t}}\ .$$

<u>Proof:</u> Same idea as in the previous limits claim. Note that we are given $p'_n = \frac{u_n}{\rho_n} p_n$. Therefore, using that $u_n = e^{\sigma\sqrt{t/n}}$ and $\rho_n = e^{rt/n}$, we have

$$p'_n = \frac{u_n}{\rho_n} p_n = \frac{e^{rt/n + \sigma\sqrt{t/n}} - 1}{e^{rt/n + \sigma\sqrt{t/n}} - e^{rt/n - \sigma\sqrt{t/n}}} = \frac{e^{rt/n + 2\sigma\sqrt{t/n}} - e^{\sigma\sqrt{t/n}}}{e^{rt/n + 2\sigma\sqrt{t/n}} - e^{rt/n}}$$

Now, we make use again of the Taylor expansion of the rightmost identity for p'_n. Observe that both 1 terms cancel out and we get

$$p'_n = \frac{rt/n + 2\sigma\sqrt{t/n} + 2\sigma^2 t/n + O(n^{-3/2}) - \left(\sigma\sqrt{t/n} + \frac{1}{2}\sigma^2 t/n + O(n^{-3/2})\right)}{rt/n + 2\sigma\sqrt{t/n} + 2\sigma^2 t/n + O(n^{-3/2}) - (rt/n + O(n^{-2}))},$$

and therefore

$$p'_n = \frac{rt/n + \sigma\sqrt{t/n} + \frac{3}{2}\sigma^2 t/n + O(n^{-3/2})}{2\sigma\sqrt{t/n} + 2\sigma^2 t/n + O(n^{-3/2})}$$

Multiplying both numerator and denominator by $\sqrt{n}$ and grouping the terms using their order of n, we get

$$p'_n = \frac{\sigma\sqrt{t} + \frac{1}{\sqrt{n}}\left(rt + \frac{3}{2}\sigma^2 t\right) + O(n^{-1})}{2\sigma\sqrt{t} + 2\sigma^2 t/\sqrt{n} + O(n^{-1})}. \tag{11}$$

Thus taking the limit as $n \to \infty$ in this expression results in

$$\lim_{n\to\infty} p'_n = \lim_{n\to\infty} \frac{\sigma\sqrt{t} + O(n^{-1/2})}{2\sigma\sqrt{t} + O(n^{-1/2})} = \frac{\sigma\sqrt{t}}{2\sigma\sqrt{t}} = \frac{1}{2}.$$

For the second limit, we use the expression in (11) and put the minus term inside the numerator so that we get

$$\sqrt{n}\left(p'_n - \frac{1}{2}\right) = \sqrt{n}\left(\frac{\frac{1}{\sqrt{n}}\left(rt + \frac{1}{2}\sigma^2 t\right) + O(n^{-1})}{2\sigma\sqrt{t} + 2\sigma^2 t/\sqrt{n} + O(n^{-1})}\right) = \frac{rt + \frac{1}{2}\sigma^2 t + O(n^{-1})}{2\sigma\sqrt{t} + O(n^{-1/2})}$$

and therefore

$$\lim_{n\to\infty} \sqrt{n}\left(p'_n - \frac{1}{2}\right) = \lim_{n\to\infty} \frac{rt + \frac{1}{2}\sigma^2 t + O(n^{-1})}{2\sigma\sqrt{t} + O(n^{-1/2})} = \frac{rt + \frac{1}{2}\sigma^2 t}{2\sigma\sqrt{t}}.$$

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<u>Claim:</u> Let p_n and p'_n be as in previous claims. Then, $1 - p'_n = (d_n/\rho_n)(1 - p_n)$.

<u>Proof:</u> Note that

$$1 - p'_n = 1 - \frac{u_n}{\rho_n} p_n = 1 - \frac{u_n}{\rho_n}\frac{\rho_n - d_n}{u_n - d_n} = 1 - \frac{u_n\rho_n - u_n d_n}{u_n\rho_n - \rho_n d_n} = \frac{d_n(\rho_n - u_n)}{\rho_n(u_n - d_n)} = \frac{d_n}{\rho_n}(1 - p_n).$$

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Theorem A.3. *If for all $y \in \mathbb{R}$, we have $\mathbb{P}(Y_n \leq y) \to \Phi(y)$ and if $x_n \to x$ for some $x \in \mathbb{R}$ as $n \to \infty$, then we also have $\mathbb{P}(Y_n \leq x_n) \to \Phi(x)$*

<u>Proof:</u> Note that $\mathbb{P}(Y_n \leq x_n) \leq \mathbb{P}(Y_n \leq x + \epsilon) = \Phi(x + \epsilon)$ for sufficiently large n. The same holds for $\mathbb{P}(Y_n \leq x_n) \geq \mathbb{P}(Y_n \leq x - \epsilon) = \Phi(x - \epsilon)$ for n' sufficiently large. Therefore

$$\Phi(x - \epsilon) \leq \mathbb{P}(Y_n \leq x_n) \leq \Phi(x + \epsilon)$$

for all $n'' = max(n, n')$.

Also, we know Φ is continuous. We can use the Squeeze theorem and we know it holds that

$$\Phi(x) \leq \mathbb{P}(Y_n \leq x_n) \leq \Phi(x)$$

for all $n'' = max(n, n')$. Therefore, we proved then $\mathbb{P}(Y_n \leq x_n) \to \Phi(x)$ as $n \to \infty$. ■

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